Jaeuk Moon

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WORK EXPERIENCE

Samsung Research (R&D center of Samsung Electronics) – Staff Engineer Sep 2025 – present

The AI Model team has successfully developed *Samsung Gauss (internal LLM) and on-device AI models for Galaxy series, including on-device LLMs for wearable health monitoring.*

❑ Responsibilities

- Conducting LLM post-training research including supervised fine-tuning (SFT) and reinforcement learning from human feedback (RLHF) to improve Samsung Gauss model alignment and deployability.
- Developing **on-device LLMs for wearable health monitoring**, applying time-series forecasting and self-supervised learning to extract health signals from noisy sensor data.
- Designed and implemented an evaluation pipeline for Samsung Health LLM candidates, combining latency and throughput measurements with rubric-based evaluation of grounding, personalization, and instruction following to support model selection.
- Developed self-supervised representations of continuous glucose monitoring (CGM) data, adapting time-series foundation models for health-related classification and postprandial glucose response prediction under limited pretraining data.
- Leading a university research collaboration on **world action models (WAMs)** for robot manipulation, covering research direction, model training and evaluation, action-conditioned future prediction, failure detection, and closed-loop correction.
- Developing GPT-as-Policy inference integrations and running VLA-driven robot control experiments in NVIDIA Isaac Sim.
- Leading independent research on LLM-based ILI (Influenza-Like Illness) forecasting, integrating multimodal inputs (numerical time-series + textual reports) with self-instructed fine-tuning to achieve explainable predictions.

Korea University Multimedia Information Laboratory – Ph.D. & Lab Manager Mar 2019 – Aug 2025

The Multimedia Information Lab (MILab) at Korea University *has successfully completed over 35 projects supported by major corporations and government agencies, published over 400 academic papers (including 100+ SCI-grade papers), registered 60+ patents, and focuses on cutting-edge AI and big data processing research in areas such as multimedia, predictive modeling, and healthcare technologies.*

❑ Responsibilities

- Led high-impact research initiatives, contributing to 18 international publications (8 first-author), 11 patent filings (10 granted), and multiple conference papers.
- As Lab Manager, streamlined research workflows, fostered interdisciplinary collaboration, and ensured successful project execution, significantly improving lab productivity.
- Managed administrative and performance tracking processes, supporting the BK21 research program and documenting progress.
- Assisted in the academic development of graduate students by managing seminars, assignments, and grading, ensuring the alignment of research objectives with academic milestones.
- Mentored junior researchers, guiding them to win 15 awards and publish 10 SCIE-indexed papers.

❑ Selected Achievements

- ***Infectious Disease Prediction Performance***: Achieved 200% of the project's publication target (6 papers published vs. 3 required) in COVID-19 projection research, collaborating with KISTI and Korea Disease Control and Prevention Agency.
- ***Developed Custom LLM Solutions***: Designed and deployed custom LLM models for industrial partners, integrating retrieval-augmented generation (RAG) to optimize industrial data workflows.
- ***ILI Forecasting:*** Developed models for influenza-like illness (ILI) forecasting under limited surveillance data, applying feature selection and region-adaptive meta-learning to improve forecasting performance by up to 30% over baseline models.
- ***B2B AI Agent Development***: Spearheaded the development of a multi-AI agent system for B2B solutions, enabling smart AI agent workflows for businesses like Incheon International Airport.
- ***Explainable AI-Based Smart Energy Platform for Energy Optimization***: Applied Explainable AI (XAI) to optimize energy usage in industrial settings, saving over $18,000 annually at a partner factory.

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- ***Achieved Record-Breaking Publications:*** Authored 8 SCIE first-author publications, the highest among lab members, establishing a strong record in academic publishing.

EDUCATION

Korea University Mar 2019 – Aug 2025

Ph.D. in Electrical and Computer Engineering (GPA 4.11/4.5)

**Doctoral Dissertation: "An AI-based Influenza-like Illness Occurrence Forecasting Scheme based on LLM" (Advisor: Dr. Eenjun Hwang)*

Korea University Mar 2015 – Feb 2019

BE in ***Electrical Engineering*** (GPA 3.26/4.5)

INDUSTRY-ACADEMIC COOPERATIVE PROJECT

Energy Cloud Technology Development Project – Researcher Jun 2019 – Dec 2023

- Collaborated with the National Research Foundation of Korea (NRF) to develop a deep learning-based probabilistic model for energy consumption prediction and simulation, focusing on scalable solutions for microgrid systems.

Infectious Disease Response Technology Development Project – Researcher Apr 2019 – Dec 2022

- Used deep learning to enhance predictive models for infectious diseases, including COVID-19, in collaboration with KHIDI (Korea Health Industry Development Institute).

Application of LLM solution and Developing AI Agent-based Services – Researcher Jan 2024 – Jun 2025

- Developed LLM-powered solutions, focusing on fine-tuning, prompt engineering, and RAG, to enhance the reliability and performance of AI systems.
- Created a self-evaluation framework to detect hallucinations in LLM-generated text, improving trust and accuracy in AI outputs.
- Designed hierarchical AI agents for optimizing AI workflows, as part of a broader initiative with an industry partner.

EXTRACURRICULAR ACTIVITIES

Graduate Seminar – Teaching Assistant Sep 2022 – Jun 2023

- Organized guest speakers and coordinated schedules for Korea University's graduate seminar course.
- Managed student attendance, evaluated reports, and calculated final grades.

Korea University BK21 Data Analysis Fellowship – Project Manager Jun 2021 – Aug 2021 / Dec 2021 – Feb 2022

- Predicted power usage of major campus buildings to prevent outages and analyzed influencing factors using explainable AI (XAI) techniques.
- Enhanced the Korea University Data Hub system by leveraging LLM-based keyword extraction to recommend research-relevant papers to graduate students.

Yulin Technology Machine Learning Seminar – Instructor Jun 2021 – Aug 2021

- Delivered seminars on machine learning applications for a small manufacturing IT company, focusing on factory operations.
- Conducted lectures and coding workshops on data preprocessing, predictive modeling types, and their practical applications.

SKILLS & LANGUAGES

- Fluent in English (OPIc AL) and Korean
- Proficient in Python & Deep Learning Frameworks (TensorFlow & PyTorch)